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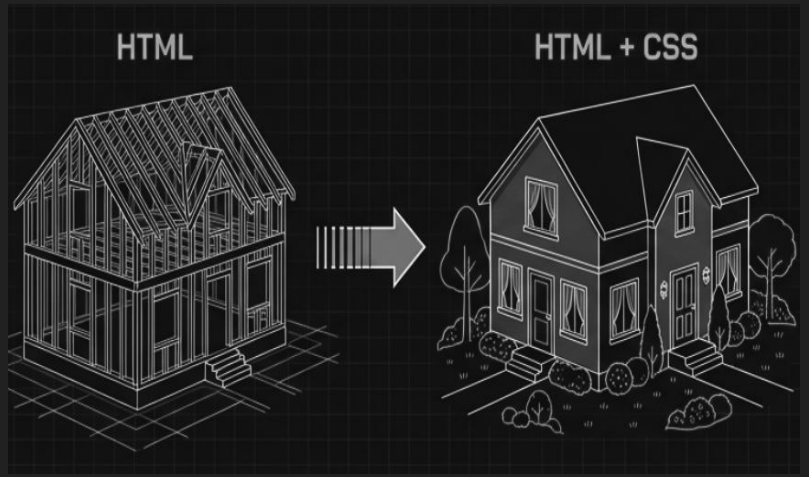
Introduction to CSS

If HTML is the raw framing and skeleton of a house the walls, the roof, the foundational bricks—-- CSS (Cascading Style Sheets) is the interior design, the paint, and the landscaping. HTML tells the browser what the content is; CSS tells the browser exactly how it should look.



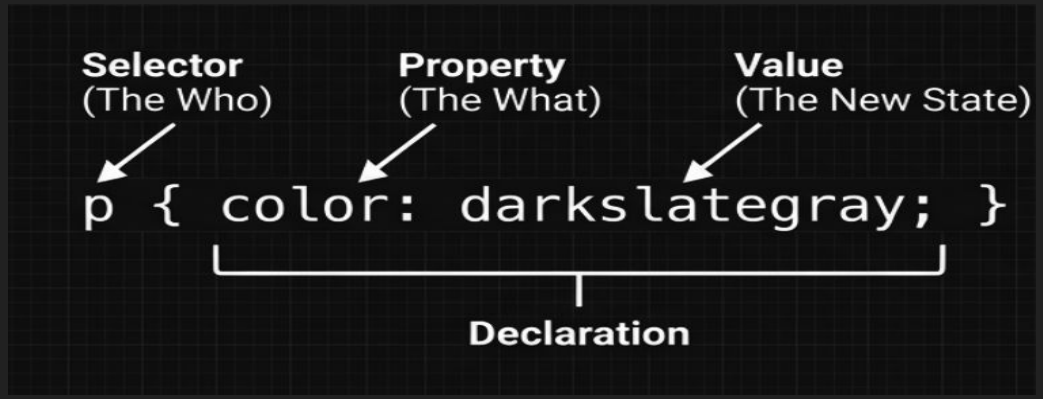

```html
<!-- HTML (The Raw Structure) -->
<h1>Welcome to My Website</h1>

<!-- CSS (The Paint & Design) -->
<style>
  h1 {
    color: royalblue;
    text-align: center;
  }
</style>
```

The Anatomy of a Declarative CSS Rule

Writing CSS is like giving a specific work order to a contractor.

Every CSS rule has three vital parts:

- **The Selector points to who you want to change.**
- **The Property specifies what feature you want to change.**
- **The Value states exactly what the new state should be.**




```
/* Selector { Property: Value; } */

p {
    color: darkslategray;
    background-color: lightgray;
}
```

Inline CSS - Messy

- **Inline CSS is written directly inside an HTML tag using the style attribute.**
- **It only affects that one specific element.**

HARD TO MAINTAIN —-- ???

```
<!-- The CSS is applied directly
inside the HTML element -->
<h2 style="color: crimson; font-
size: 24px;">
  This is an urgent warning!
</h2>
```

Internal CSS - Single Room Decoration

- Internal CSS is placed inside a <style> tag, which lives inside the <head> of your HTML document .
- It can style anything inside that specific HTML page, but it cannot share those styles with any other pages on your website.

```
<head>
  <title>About Us</title>
  <style>
    body {
      background-color: #f4f4f4;
    }
    p {
      color: #333333;
    }
  </style>
</head>
```

External CSS - Master Blueprint

- **This is the modern industry standard for web engineering. You write all your CSS in a completely separate file (ending in .css) and link it to your HTML using the <link> tag.**
- **One single CSS file can control the design of thousands of HTML pages simultaneously.**

HTML

```
<link rel="stylesheet" href="styles.css">
```

```
<!-- Inside your HTML <head> -->
<link rel="stylesheet"
href="styles.css">


/* Inside styles.css (a totally
separate file) */
nav {
  background-color: black;
  color: white;
}
```

Cascade Resolves Design Conflicts

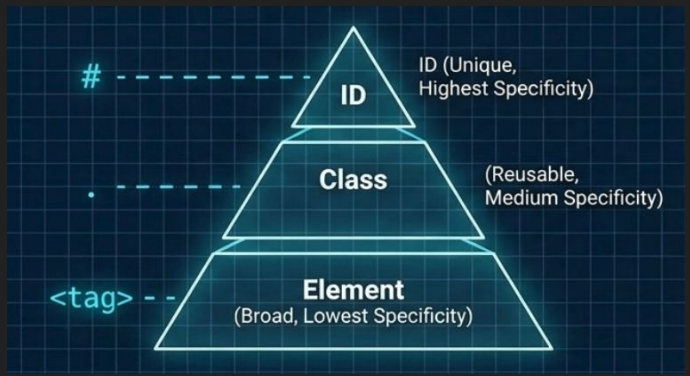
- Why is it called Cascading Style Sheets? Because styles flow downward like a waterfall.
- If multiple rules target the exact same element, the browser needs a way to decide which one wins.
- Generally, the rule with the highest specificity wins. If specificity is an exact tie, the rule written last (closest to the bottom of the cascade) overrides the earlier ones.

```
/* In this scenario, the button
will be GREEN. */
/* The browser reads top-to-bottom,
so the last rule wins a tie. */

button {
  background-color: blue;
}

button {
  background-color: green;
}
```

Elements: Classes vs. IDs

To target elements effectively, we use different selector types:

- **Element selectors target raw HTML tags broadly.**
- **Class selectors (starting with a .) are reusable and carry moderate specificity—they are the absolute workhorse of modern CSS.**
- **ID selectors (starting with a #) carry high specificity but must be completely unique per page.**



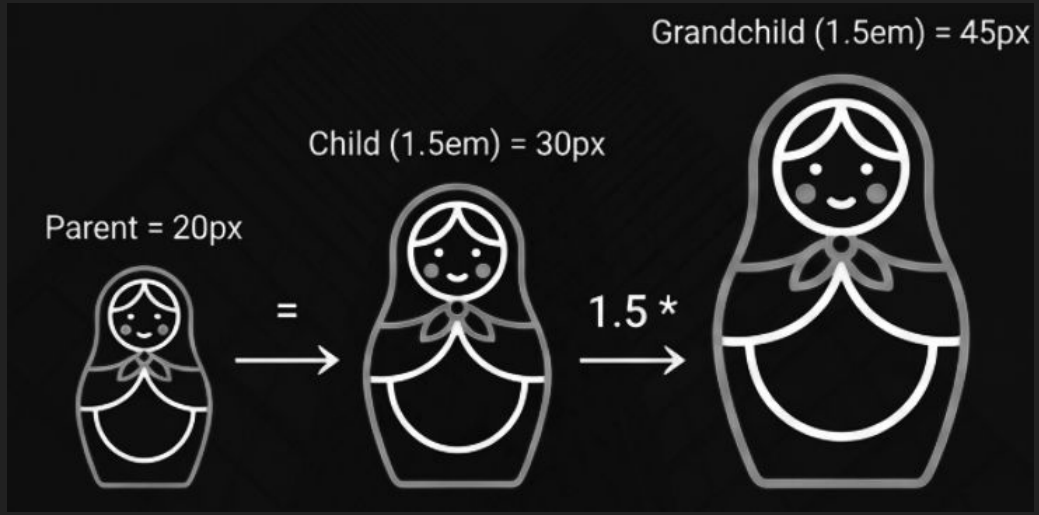

```css
/* Element Selector (Broadest) */
p { line-height: 1.5; }

/* Class Selector (Reusable Workhorse) */
.highlight-box { border: 2px solid yellow; }

/* ID Selector (Unique, High
#main-navigation {
  background: black;
}
```

Fixed Sizes: Using Pixels (px)

- When sizing elements or text, we must choose a unit of measurement.
- A Pixel (px) is an Absolute Unit. It is like measuring something with a rigid wooden ruler. If you explicitly tell a box to be 300px wide, it will stay exactly 300px wide regardless of the user's screen size or default browser settings.
- Pixels are highly predictable for borders, but poor choices for accessible text sizing.

Flexible Sizes: Sizing with "em"

Relative units act like a flexible tape measure - they adapt based on their surroundings. The em unit is relative to the font size of its direct parent element.




```html
<div class="parent">
  <p class="child">I am sized
relative to my parent.</p>
</div>
```

```css
.parent {
  font-size: 20px;
}

.child {
  /* 1.5 * 20px = 30px */
  font-size: 1.5em;
}
```

Consistent Flexible Sizes: Sizing with "rem"

- To fix the compounding problem of em, professional engineers use rem (Root EM).
- A rem is always relative to the root font size of the entire HTML document (usually 16px), no matter how deeply it is nested.
- It is the gold standard for accessible typography. If a user increases their browser's default font size, your entire site scales perfectly in unison.

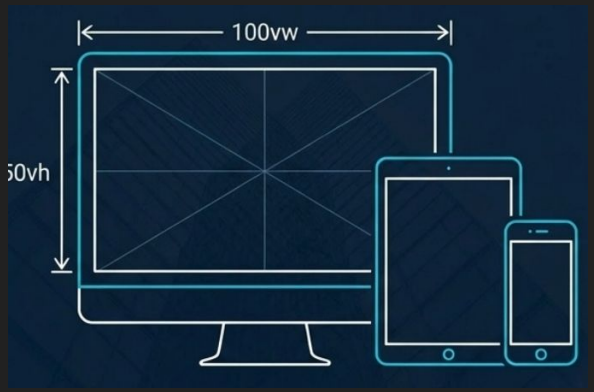
```
html {
  /* The global root size */
  font-size: 16px;
}

h1 {
  /* 3 * 16px = 48px, regardless of HTML nesting */
  font-size: 3rem;
}
```

Screen-Based Sizes: Making Layouts Fit Any Device

When building layouts that must seamlessly adapt to phones, tablets, and desktops, we use units relative to the browser window itself (the Viewport).

- **1vw is exactly 1% of the screen's total width.**
- **1vh is 1% of the screen's total height.**

Percentages (%) are similar, but they are relative to the parent container's size rather than the viewport.




```css
.hero-banner {
  /* Spans 100% of the browser's width */
  width: 100vw;
  /* Takes up exactly half of the browser's height */
  height: 50vh;
}

.child-box {
  /* Takes up half the width of the .hero-banner */
  width: 50%;
}
```

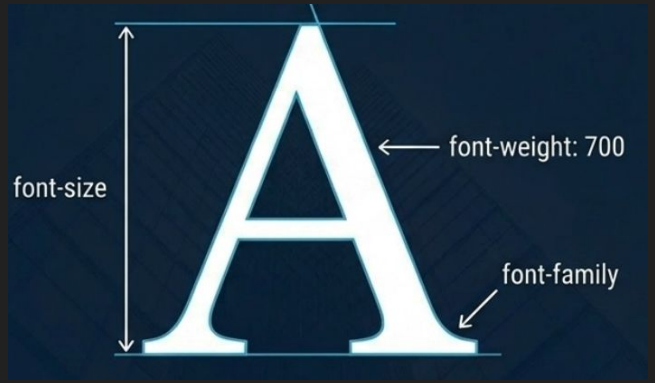
Quick Guide: Choosing the Right Size Unit

Unit	Relative to...	Best Use Case	Accessibility
px	Nothing (Rigid)	Borders, tiny details	Poor
rem	Root Document	All Typography	Excellent
em	Direct Parent	Proportional Padding	Good (Careful)
vh / vw / %	Viewport / Parent Box	Macro Layouts	Neutral

```css
.perfect-button {
  border: 2px solid black; /* Rigid px */
  font-size: 1.2rem;   /* Accessible rem */
  padding: 0.5em 1em; /* Proportional em */
  width: 100%;        /* Responsive % */
}
```

Text Basics: Font Style and Thickness

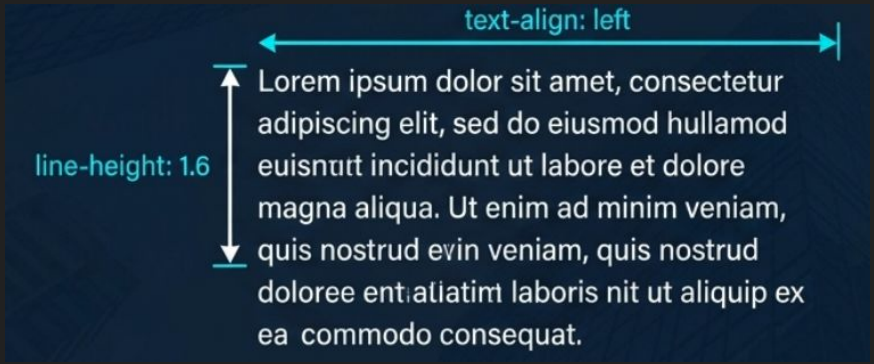
- Typography accounts for 90% of web design.
- The three most critical properties are font-family (the typeface you select), font-size (how large it is), and font-weight (how thick or bold the letters are).
- Always provide a generic fallback font (like sans-serif) to prevent layout crashes if your primary font fails to load.




```
p {
  /* Arial is primary, sans-serif is the safety backup */
  font-family: "Arial", sans-serif;
  font-size: 1.1rem;

  /* 400 is standard normal text, 700 is bold */
  font-weight: 700;

}
```

Web Text: Using Google Fonts and Alignment

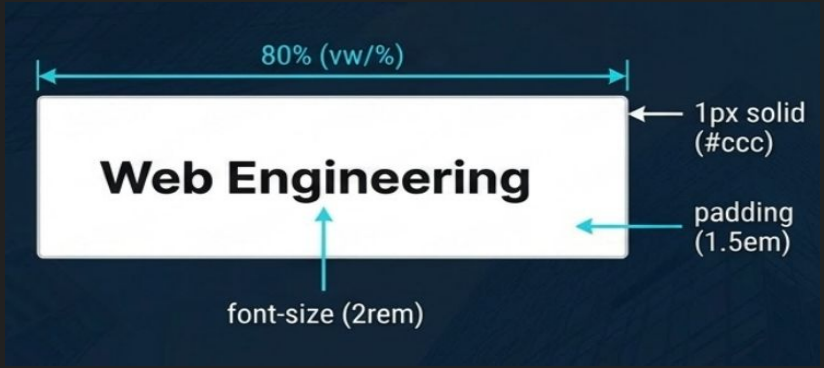
- **You are not limited to local fonts. The Google Fonts API is a massive library for web typography.**
- **You fetch these font files via an @import rule placed at the very top of your CSS file.**
- **Once imported, you refine the reading experience using line-height to create vertical breathing room, and text-align to structure the paragraph layout.**




```css
/* 1. Import at the very top of your CSS file */
@import url('https://fonts.googleapis.com/css2?family=
Roboto:wght@400;700&display=swap');

/* 2. Apply and refine */
article {
  font-family: 'Roboto', sans-serif;
  line-height: 1.6; /* Crucial vertical breathing room */
  text-align: left;
}
```

Putting It All Together: Building a Real Component

- Let's bring all our tools together into a final blueprint.
- We will link an external stylesheet, import a custom web font, use rem for accessible text sizing, em for proportional spacing, and absolute px for borders.
- This code block represents exactly what professional, manual CSS engineering looks like in practice.




```html
<link rel="stylesheet" href="main.css">

<div class="card">
  <h2>Web Engineering</h2>
</div>
```

```css
@import url('https://fonts.googleapis.com/css2?family=Inter&display=swap');

.card {
  font-family: 'Inter', sans-serif;
  width: 80%; /* Responsive layout */
  border: 1px solid #ccc; /* Rigid detail */
  padding: 1.5em; /* Proportional spacing */
}

.card h2 {
  font-size: 2rem; /* Accessible typography */
  line-height: 1.2;
}
```

THANK YOU!

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